Signed transfer invisible to spectral energy statistics: exact Gaussian triad diagnostics

Project Stardust

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Abstract

We reconstruct a finite real solenoidal Fourier family and its exact weighted sector-transfer ratio. Randomizing one signed amplitude by a standard Gaussian gives a nonzero odd transfer observable. Every even observation has zero conditional mean transfer, although weighted energy determines its square and hence its strictly positive conditional variance. Positive bounded changes of probability density preserve the entire joint law of energy and concentration while reversing mean signed transfer. Complete proofs distinguish equality of observed laws from equality of latent laws. We also derive, explicitly as classical consequences, the biased conditional means and a two-law prediction-risk identity. The value is an exact diagnostic and formalization companion: phase dependence and symmetry-based information loss are classical, and priority for this particular package remains unresolved. No turbulent evolution, invariant Navier–Stokes measure or failure of a dynamically constrained closure is asserted.

1 What is observed and what is tested

Spectral energy is quadratic in Fourier coefficients, whereas a nonlinear transfer is cubic. Sign information can therefore disappear from energy statistics. We make this statement exact on an explicitly specified family, partition and probability law, including all normalizing denominators. The signed parameter below changes relative signs of imaginary Fourier coefficients; it is not an angle uniformly distributed on a circle.

The source calls $\mathcal{E}$ an energy. It is the weighted quantity $\sum_k |k|^2 |\widehat{u}(k)|^2$, proportional to enstrophy on this solenoidal family. It is not the unweighted kinetic energy. Our central observation is $S = (\mathcal{E}, \mathcal{C})$, with the concentration $\mathcal{C}$ defined below. Its entire probability law will be preserved, not just one or two moments.

All finite formulas are evaluated on static fields. We define a transfer diagnostic by its displayed convolution formula. We do not assume that the six retained modes remain closed under Euler or Navier–Stokes evolution. Moffatt's primary analysis [1], introduction and Section 2, makes precisely this distinction between triad truncation and exact fluid evolution. None of his evolution results is needed for the finite identities here.

2 A real periodic triad and its complete finite transfer

Let $e_1 = (1,0,0)$, $e_2 = (0,1,0)$ and $K = \{\pm e_1, \pm e_2, \pm(e_1+e_2)\} \subset \mathbb{Z}^3$. For real a, b, c define a real vector polarization V by

$$V(e_1) = (0,a,0), \quad V(e_2) = (-b,0,0), \quad V(e_1+e_2) = (-c,c,0), \quad V(-k) = -V(k),$$

Set $V = 0$ off K. Use complex Fourier coefficients $iV(k)$ on the period-2π three-torus, so that

$$u(x) = \sum_{k \in K} iV(k) e^{ik \cdot x}.$$

Pairing opposite modes gives the actual field

$$u_{a,b,c}(x) = \big(2b \sin x_2 + 2c \sin(x_1 + x_2), -2a \sin x_1 - 2c \sin(x_1 + x_2), 0\big). \tag{1}$$

Thus u is real, smooth and periodic. Each $k \cdot V(k) = 0$; equivalently its two displayed divergence terms are $2c \cos(x_1 + x_2)$ and $-2c \cos(x_1 + x_2)$. This proves incompressibility without a pressure assumption. The field vanishes identically only if $a = b = c = 0$: evaluate its components at $(\pi/2, 0, 0)$ and $(0, \pi/2, 0)$ to obtain first $c = 0$ and then $a = b = 0$.

For $k \in K$ define

$$W_k = |k|^2 |V(k)|^2, \qquad T_k = 2|k|^2 \sum_{\substack{p,q \in K \\ p+q=k}} (q \cdot V(p))(V(k) \cdot V(q)). \tag{2}$$

Because V is zero off K, the sum agrees with the unrestricted convolution at each retained output frequency: an outside input has a zero vector factor. This statement does not assert vanishing at other output frequencies or invariance of the finite support under evolution.

Lemma 2.1. *In the order $e_1, e_2, e_1 + e_2, -e_1, -e_2, -e_1 - e_2$, the weights and transfers are*

$$(W_k) = (a^2, b^2, 4c^2, a^2, b^2, 4c^2), \quad (T_k) = (2abc, -2abc, 0, 2abc, -2abc, 0).$$

In particular $\sum T_k = 0$ and $\mathcal{E} := \sum W_k = 2a^2 + 2b^2 + 8c^2$.

Proof. The weight formula follows from $|e_1|^2 = |e_2|^2 = 1$ and $|e_1 + e_2|^2 = 2$. For output e_1, the only ordered pairs are $(e_1 + e_2, -e_2)$ and $(-e_2, e_1 + e_2)$. The first summand vanishes because $V(e_1) \cdot V(-e_2) = 0$; the second is $b(ac) = abc$. The prefactor is two, giving $2abc$. For output e_2, the pairs are $(e_1 + e_2, -e_1)$ and $(-e_1, e_1 + e_2)$. The first has a zero vector inner product and the second is $(-a)(bc) = -abc$, giving $-2abc$. For output $e_1 + e_2$, the pairs (e_1, e_2) and (e_2, e_1) contribute abc and $-abc$. Their sum is zero. Replacing (k, p, q) by $(-k, -p, -q)$ leaves each summand unchanged: both dot products contain two sign reversals. This proves the three opposite-output values and completes every possible pair. Summing the two rows gives the last assertions. □

3 An actual directional partition and the ratio

We specify the source's 32 labels rather than infer a tessellation from the six sample directions. For a direction (x, y, z), assign a quadrant $Q \in \{0, 1, 2, 3\}$ by

$$Q = \begin{cases} 0 & y \geq 0,\ x \geq 0, \\ 1 & y \geq 0,\ x < 0, \\ 2 & y < 0,\ x < 0, \\ 3 & y < 0,\ x \geq 0. \end{cases}$$

Assign a bin $B \in \{0, \dots, 7\}$ by the first satisfied test in the following ordered list, and assign $B = 7$ if none is satisfied:

$$4|y| < |x|, \quad 2|y| < |x|, \quad 4|y| < 3|x|, \quad |y| < |x|,$$
$$3|y| < 4|x|, \quad |y| < 2|x|, \quad |y| < 4|x|.$$

The sector is $s = 8Q + B$. This rule assigns each point of the unit sphere exactly one label and is invariant under positive scaling. Every label is attained: in quadrant zero take $(1, r, 0)$ with

$$r = 1/8,\ 3/8,\ 5/8,\ 7/8,\ 7/6,\ 3/2,\ 3,\ 5$$

for the eight bins, change the signs to enter the other quadrants, and normalize to unit length. Normalization preserves every strict comparison and quadrant. Thus these are disjoint nonempty cells covering the sphere; no smooth-boundary or equal-area claim is needed. On the six frequencies above their labels are

$$(s(k)) = (0, 7, 4, 8, 31, 20).$$

They are distinct, even though the general rule has 32 cells.

Put $E_j = \sum_{s(k)=j} W_k$, $F_j = \sum_{s(k)=j} T_k$ and, for $\mathcal{E} > 0$, define

$$p_j = E_j/\mathcal{E}, \qquad \mathcal{C} = \sum_{j=0}^{31} p_j^2,$$
$$\Sigma = \frac{2}{\mathcal{E}} \sum_{j=0}^{31} p_j \left(F_j - p_j \sum_k T_k \right), \qquad R = -\frac{\Sigma}{\mathcal{C} - 1/32}. \tag{3}$$

This precise formula defines the diagnostic being tested. Its denominator and its sign will be verified on actual amplitudes.

Proposition 3.1. *For $(a, b, c) \neq 0$ let $E = 2a^2 + 2b^2 + 8c^2$ and $A = a^4 + b^4 + 16c^4$. Then*

$$\mathcal{C} = \frac{2A}{E^2} \geq \frac{1}{6} > \frac{1}{32}, \qquad \Sigma = \frac{8abc(a^2 - b^2)}{E^2}, \qquad R = \frac{-8abc(a^2 - b^2)}{2A - E^2/32}. \tag{4}$$

The denominator is strictly positive. Swapping a, b preserves $E, \mathcal{C}$ and reverses R. At $(2, 1, 1)$ and $(1, 2, 1)$ the ratios are $-128/149$ and $128/149$. At $(2, 2, 1)$ the ratio is zero although $T_{e_1} = 8$.

Proof. The six displayed sector weights are the six entries of Lemma 2.1. Their squared sum is $2A$, proving the concentration formula. Cauchy–Schwarz for these six weights gives $E^2 \leq 6(2A)$, hence the displayed lower bound. Because $E > 0$, $2A - E^2/32 = E^2(\mathcal{C} - 1/32) > 0$. The total transfer vanishes, while

$$\sum_j E_j F_j = 2a^2(2abc) + 2b^2(-2abc) = 4abc(a^2 - b^2).$$

Substitution in (3) gives Σ and then R. The numerator is antisymmetric in a, b and its denominator symmetric. At $(2, 1, 1)$, $E = 18$, $A = 33$ and $2A - E^2/32 = 447/8$, giving $-48/(447/8) = -128/149$. At $(2, 2, 1)$, $E = 24$, $\mathcal{C} = 1/6$, and the imbalance $abc(a^2 - b^2)$ vanishes, while the table gives $T_{e_1} = 8$. Thus zero normalized rate does not mean that every individual transfer vanishes. All assertions follow from the actual finite table. □

4 Gaussian signed amplitudes and exact invisible fluctuations

Now fix $a = 2, b = 1$ and let C have standard Gaussian law γ. To avoid confusion between a random variable and concentration, write $r(c) = R(2, 1, c)$ and $d(c) = 30c^4 - 5c^2 + 247/8$. Proposition 3.1 gives

$$r(c) = -\frac{48c}{d(c)}, \qquad d(c) = 30(c^2 - 1/12)^2 + 92/3 \geq 92/3. \tag{5}$$

Consequently r is continuous and odd, vanishes only at zero and has the sign opposite to c. In particular

$$|r(c)| \leq \frac{36}{23}|c|.$$

This proves $r \in L^2(\gamma)$ and integrability of its square and all products with L^2 predictors by Cauchy–Schwarz. Gaussian symmetry gives $\mathbb{E}_\gamma r = 0$. Since a nondegenerate Gaussian has no atom at zero, $r(C)^2 > 0$ almost surely, so

$$0 < v := \mathbb{E}_\gamma r(C)^2 < \infty. \tag{6}$$

The exact integral need not be evaluated to prove this strict inequality. Likewise $r(c)c = -48c^2/d(c) < 0$ off zero, and its integral is finite and strictly negative.

Theorem 4.1. *If $F : \mathbb{R} \to \mathsf{Y}$ is any measurable even observation and $\mathcal{A} = \sigma(F(C))$, then*

$$\mathbb{E}_\gamma[r(C) \mid \mathcal{A}] = 0 \quad \textit{almost surely.}$$

For every $\mathcal{A}$-measurable $g \in L^2(\gamma)$,

$$\mathbb{E}_\gamma(r - g)^2 = v + \mathbb{E}_\gamma g^2. \tag{7}$$

In particular zero is the unique optimal L^2 predictor, and its error is the strictly positive number v.

Proof. Every event $A \in \sigma(F)$ is invariant under $c \mapsto -c$: this holds for preimages of measurable sets and the collection of symmetric sets is a sigma algebra. Hence $1_A(c)r(c)$ is odd and integrable, and Gaussian symmetry gives $\int_A r \, d\gamma = 0$. The zero function is $\mathcal{A}$-measurable and satisfies these integral identities, so it is a conditional expectation of r. For a bounded $\mathcal{A}$-measurable g, the identities first give $\mathbb{E}rg = 0$ for simple functions and then by bounded approximation. For general L^2 g, truncate it and use Cauchy–Schwarz to pass to the limit. Expanding the square proves (7). Its extra term is nonnegative, with equality exactly when $g = 0$ in L^2. □

Our actual observations are

$$E(c) = 10 + 8c^2, \qquad H(c) = \frac{2(17 + 16c^4)}{(10 + 8c^2)^2}, \qquad S(c) = (E(c), H(c)). \tag{8}$$

They are continuous and even. Moreover E recovers $|c|$ by $|c| = \sqrt{(E - 10)/8}$ on its range $E \geq 10$. Thus $\sigma(S) = \sigma(E) = \sigma(|C|)$ on this experiment. This identifies exactly which information is retained, rather than merely showing that some statistic is even.

Corollary 4.2. *Define, for $e \geq 10$,*

$$J(e) = \frac{2304((e-10)/8)}{(30((e-10)/8)^2 - 5((e-10)/8) + 247/8)^2}.$$

Then $r(c)^2 = J(E(c))$ and

$$\mathrm{Var}_\gamma(r(C) \mid S) = J(E(C)) > 0 \quad \textit{almost surely.}$$

Proof. Substitute $(E(c) - 10)/8 = c^2$ in (5) squared. The square is integrable and already measurable in E, so its conditional expectation is itself. The conditional mean is zero by Theorem 4.1. Conditional variance is their difference, and positivity follows from $C \neq 0$ almost surely. □

The same parity calculation determines the first nonzero Hermite index. Let H_n be the probabilists' polynomials with $H_0 = 1$, $H_1 = c$ and $H_{n+1} = cH_n - nH_{n-1}$. Induction gives $H_n(-c) = (-1)^n H_n(c)$. Polynomial Gaussian moments and (5) imply integrability of $H_n r$. Therefore

$$\mathbb{E}_\gamma[H_{2n}(C)r(C)] = 0, \qquad \mathbb{E}_\gamma[H_1(C)r(C)] = \mathbb{E}_\gamma[Cr(C)] < 0.$$

The mean coefficient is zero and the degree-one coefficient is nonzero: the Hermite rank is exactly one. This does not contradict invisible prediction from S, which observes only the even coordinate $|C|$. It is a parity statement about a different observation space.

5 Identical observed laws with opposite mean transfer

Put

$$b(c) = \frac{c}{1+|c|}, \qquad d\gamma_\varepsilon(c) = (1 + \varepsilon b(c))\, d\gamma(c), \qquad |\varepsilon| \leq 1.$$

The use of b here is local to this probability section; the fixed second triad amplitude remains one. Since b is continuous, odd and $|b(c)| < 1$, the density is strictly positive at every finite c, bounded above by two, and integrates to one. Thus these are genuine probability laws, equivalent to γ, including the endpoints $\varepsilon = \pm 1$. All γ-integrable functions remain integrable under them.

Theorem 5.1. *For every measurable even map F, the full pushforward laws satisfy $F_\# \gamma_\varepsilon = F_\# \gamma$. In particular*

$$S_\# \gamma_1 = S_\# \gamma_{-1} = S_\# \gamma.$$

Nevertheless, for the finite positive constant

$$m = -\mathbb{E}_\gamma[b(C)r(C)] > 0,$$

one has $\mathbb{E}_{\gamma_\varepsilon} r = -\varepsilon m$. Thus γ_1 and γ_{-1} are distinct and give opposite strict mean transfer while their joint observed laws agree exactly.

Proof. For a measurable observation event D, its preimage $A = F^{-1}(D)$ is symmetric. The integrable function $1_A b$ is odd, so

$$\gamma_\varepsilon(A) = \int_A (1 + \varepsilon b)\, d\gamma = \gamma(A).$$

This is equality on every measurable set of the observation space, which proves equality of the pushforwards, not just moment matching. The product br is integrable because b is bounded and r is integrable. Formula (5) gives

$$b(c)r(c) = -\frac{48c^2}{(1+|c|)d(c)} < 0 \qquad (c \neq 0).$$

Its negative is positive almost surely and integrable, so $0 < m < \infty$. Finally integrate $r(1+\varepsilon b)$ and use $\mathbb{E}_\gamma r = 0$. Different means of one integrable observable prove that the two endpoint laws are distinct. □

The deterministic case already has the same sign reversal at $c = 1$ and $c = -1$, with unchanged observed values. The probability statement is stronger: it preserves the distribution of every even measurable statistic under strictly positive changes of density, rather than pairing only two deterministic samples. It excludes determining this mean transfer from the law of S alone over the displayed family of experiments. It does not exclude a closure which additionally specifies a conditional phase law or dynamical assumptions.

6 Conditional effects of the bias and a common-risk identity

The following consequences are proved here from the mapped formulas; they are not attributed to additional named Lean declarations. They clarify how conditional mean and uncertainty change even when the observation law is completely unchanged.

For $r_0 = |c|$, the products $b(c)r(c)$ and $r(c)^2$ are even and are measurable in $E(c)$. The tilted conditional expectations are

$$\mathbb{E}_{\gamma_\varepsilon}[r \mid S] = \varepsilon br, \qquad \mathbb{E}_{\gamma_\varepsilon}[r^2 \mid S] = r^2. \tag{9}$$

To prove the first identity, test against any bounded $\sigma(S)$-measurable g. Such a function is even, and

$$\mathbb{E}_{\gamma_\varepsilon}[rg] = \mathbb{E}_\gamma[rg] + \varepsilon \mathbb{E}_\gamma[brg] = \varepsilon \mathbb{E}_\gamma[brg].$$

Since brg is even, its integral under γ_ε equals the one under γ by Theorem 5.1. The right side is therefore $\mathbb{E}_{\gamma_\varepsilon}[(\varepsilon br)g]$. All terms are integrable, giving the defining conditional identities. For the second formula, r^2 is already integrable and observed. It follows that

$$\mathrm{Var}_{\gamma_\varepsilon}(r \mid S) = r^2(1 - \varepsilon^2 b^2) > 0 \quad \text{almost surely.} \tag{10}$$

Here strict positivity also holds at $|\varepsilon| = 1$, because $|b(c)| < 1$ for every finite c and $r(c) \neq 0$ off zero. Equivalence of the tilted Gaussian laws preserves that null exception.

There is also a prediction obstruction if one must use the same observation rule without being told which endpoint law applies. Let $g = \varphi(S)$ with $\mathbb{E}_\gamma g^2 < \infty$, and write $\mathcal{R}_\pm(g) = \mathbb{E}_{\gamma_{\pm 1}}(r-g)^2$. Bounded densities ensure these errors are finite. Averaging the two densities cancels the odd bias exactly, so

$$\frac{\mathcal{R}_+(g) + \mathcal{R}_-(g)}{2} = \mathbb{E}_\gamma(r-g)^2 = v + \mathbb{E}_\gamma g^2 \geq v. \tag{11}$$

Thus at least one of the two errors is at least v. The common rule $g = 0$ has error exactly v under both laws, because r^2 is even. Consequently the minimum over common rules of $\max\{\mathcal{R}_+(g), \mathcal{R}_-(g)\}$ is exactly v. This is an elementary two-experiment consequence, with a complete proof in (11); no new minimax principle is claimed. The best rule when ε is known is instead εbr, as (9) states. The common-rule obstruction must not be read as its failure.

7 Primary comparison and formal correspondence

Fourier triad transfer and phase dependence are established topics. Moffatt [1] was inspected in the primary PDF through the introduction and the two-dimensional representation in Section 2; its distinction between instantaneous triad data and exact evolution is retained here. Murray–Bustamante [2] was checked at primary abstract level for phase dynamics with amplitudes held fixed in a Burgers model. Benavides–Bustamante [3], submitted May 2026, was checked at primary abstract level for a stochastic phase closure in two-dimensional turbulence. Its statistical and dynamical assumptions are not hypotheses of our static experiment. Our nonidentifiability example is not presented as a counterexample to either paper or to a fully specified turbulent closure.

Parity orthogonality, conditional expectation as L^2 projection and bounded changes of density are classical. The contribution is a fully reproducible exact family combining the actual partition, solenoidal representative, signed transfer denominator and full-law information loss. Priority of the particular formulas and combined formalization remains unresolved after this bounded comparison.

All formal sources are frozen at commit

`bab688bb6140cd6316c80680095a914ea0d346db.`

The accompanying theorem map records exact declaration names, headers, lines, file hashes and permanent links. The main modules are:

Actual finite family	`ParameterizedSectorTriad`
Directional cells	`AngularTriadPartition`
Real representative	`ParameterizedTriadRepresentative`
Gaussian rate and signs	`GaussianTriadPhase`
Hermite parity/rank	`GaussianTriadHermite`
Positive biased laws	`GaussianTriadBias`
Conditional mean/variance	`GaussianTriadConditional`

The finite table is proved by enumerating every interacting pair. The concentration bound is reproved by six-term Cauchy–Schwarz. Equality of the sigma fields $\sigma(S) = \sigma(|C|)$, strict equivalence at the endpoint tilted laws, the biased conditional formulas and the common-risk identity are explicit classical deductions proved here. The source conditional variance theorem concerns the original symmetric Gaussian; the tilted formula is not misattributed to it.

Original build/axiom evidence and source hashes were checked at the frozen baseline; no new Lean build or mathematical source change is claimed. Exact finite algebra checks independently reconstruct the triad table, labels, denominator and scalar identities. They do not replace the printed measure-theoretic proofs with Gaussian sampling or numerical quadrature.

The scope is static finite Fourier geometry and an actual scalar Gaussian experiment. No invariant fluid probability measure, stochastic turbulence model, continuum closure theorem, global regularity conclusion or singularity mechanism is established.

References

[1] H. K. Moffatt, Note on the triad interactions of homogeneous turbulence, *J. Fluid Mech.* **741** (2014), R3. doi:10.1017/jfm.2013.637. Primary PDF: `https://www.damtp.cam.ac.uk/user/hkm2/PDFs/Moffatt2014a.pdf`.

[2] B. P. Murray and M. D. Bustamante, Energy flux enhancement, intermittency and turbulence via Fourier triad phase dynamics in 1D Burgers equation, primary preprint arXiv:1705.08960v2 (2018). `https://arxiv.org/abs/1705.08960`. Abstract-level comparison only.

[3] S. J. Benavides and M. D. Bustamante, Triad phase dynamics determine cascade direction in two-dimensional turbulence, primary preprint arXiv:2605.03049v1 (2026). `https://arxiv.org/abs/2605.03049`. Abstract-level comparison only.